

BMRK557_Module 2

Literature Review and Technical Reading

Literature Review and Technical Reading:

New and Existing Knowledge, Analysis and Synthesis of Prior Art
Bibliographic Databases, Web of Science, Google and Google Scholar,
Effective Search: The Way Forward Introduction to Technical Reading
Conceptualizing Research, Critical and Creative Reading, Taking Notes
While Reading, Reading Mathematics and Algorithms, Reading a
Datasheet.

Attributions and Citations:

Giving Credit Wherever Due, Citations: Functions and Attributes, Impact
of Title and Keywords on Citations, Knowledge Flow through Citation,
Citing Datasets, Styles for Citations, Acknowledgments and Attributions,
What Should Be Acknowledged, Acknowledgments in, Books
Dissertations, Dedication or Acknowledgments.

Literature Review

- The term “literature” is commonly used to refer to the body of written works on a particular subject or within a specific field of study.
- A literature review, is a systematic and critical analysis of existing scholarly works, literature, and research relevant to a specific research topic.
- The primary objectives of literature review are :
 - **Identification of Vaguely Known Problem:** To know the use of content, ideas, or approaches in the literature to correctly identify the problem that is not clearly known before delving into existing research.
 - **Advocating a Specific Approach:** The literature review enables researchers to advocate for a specific approach or methodology in understanding and addressing the identified problem.
 - **Assessing Methods Used:** Researchers evaluate and assess the methods used in the literature to address similar problems.

Literature Review

- **Ensuring Contribution of Something New and Innovative:** The literature review helps researchers in clearly understanding that their proposed research (the research to be undertaken) will contribute something novel and innovative.
- **Evaluating the Quality of a Literature Review:** The quality of such a review can be determined by evaluating whether it includes the appropriate breadth and depth of the area under study, clarity, rigor, consistency, and effective analysis.
- **New and Existing Knowledge**
- New knowledge in research can only be interpreted within the context of what is already known, and cannot exist without the foundation of existing knowledge.
- **Foundation of Existing Knowledge:** New knowledge is built upon the foundation of existing knowledge. Understanding what is already known is essential or very important in the context of academic writing or research and to support and understand new findings.

New and Existing Knowledge

- **Constructing the Foundation:**

Constructing a strong foundation involves reading and surveying literature, both historical and recent. Existing knowledge provides context, significance, originality, and tools necessary for new research.

- **Sources of Existing Knowledge:**

Normally, one finds this knowledge by reading and surveying the literature in the field that was established long ago, as well as more recent knowledge, which is always changing.

- **(i) Textbooks:** Literature survey involves reading textbooks on one's topic for established knowledge. Textbooks serve as a starting point for understanding basics. Reading a textbook is not too difficult because it is written as a teaching instrument. The author of the textbook normally starts from the basics and takes the reader through everything needed to understand that topic.

- **(ii) Research Papers:** Research papers contain newer work but assume prior knowledge. Normally, the goal of a research paper is to present a small piece of new knowledge. A research paper contributes newer work but assumes a certain level of prior knowledge in the field from the reader.

Steps of Literature survey

- A good literature survey is typically a two-step process as enumerated below:
- Identify the major topics or subtopics or concepts relevant to the subject under consideration.
- Place the citation of the relevant source (article/patent/website/data, etc.) in the correct category of the concept/topic/subtopic (with the help of a, for example).

Analysis and Synthesis of Prior ART

- After collecting the sources, usually articles, intended to be used in the literature review, the researcher is ready to break down each article and identify the useful content in it, and then synthesize the collection of articles. Breaking Down Each Article:
- A researcher should analyze the relevant information
 - (i) Understanding the Hypothesis : Clearly comprehend the main hypothesis or research question addressed in each article.
 - (ii) Understanding Models and Experimental Conditions: Gain insight into the models and experimental setups used in each study.
 - (iii) Making Connections: Identify commonalities, patterns, or trends across the articles.
 - (iv) Comparing and Contrasting: Highlight similarities and differences among the articles.
 - (v) Finding Strong Points and Loopholes: Evaluate the strengths and weaknesses of each article, acknowledging areas of robust evidence and potential limitations.

Literature Survey Grid

Topic/Source	Source 1	Source 2	Source 3		Source M
Topic 1	----				----
Topic 2		-----	---		
.....					
Topic N	----	-----			-----

Bibliographic Databases

- Bibliographic databases are referred to as “abstracting and indexing services.”
- **Purpose:** They play a crucial role in collecting citation-related information and abstracts of research articles from scholarly literature.
- **Accessibility:** The gathered information is made available through search functionalities, aiding researchers in retrieving relevant literature efficiently.
- **Example Databases:** IEEE Xplore, web of science, Scopus, Google scholar, Pubmed, Science Direct etc.

Web of Science

- Web of Science (formerly known as ISI or Thomson Reuters) includes multiple databases, as well as specialized tools. It is a good search tool for scholarly materials requiring institutional license and allows the researcher to search in a particular topic of interest.
- Search criteria include the fields that are available in drop down menu such as title, topic, author, address, etc. The tool also allows sorting by number of citations (highest to lowest), publication date etc.

The screenshot shows the Master Journal List interface. At the top, there's a navigation bar with links like 'Search Journals', 'Match Manuscript', 'Downloads', and 'Help Center'. A search bar contains 'journal of imaging', and a 'Search' button is next to it. To the right, a 'Sort By' dropdown is set to 'Relevancy'. Below the search bar, 'Active Filters' are listed: 'SCIENCE CITATION INDEX EXPANDED (SCIE)', 'SOCIAL SCIENCES CITATION INDEX (SSCI)', 'ARTS & HUMANITIES CITATION INDEX (AHCI)', and 'EMERGING SOURCES CITATION INDEX (ESCI)'. The 'Search Results' section indicates 'Found 341 results (Page 1)' and includes a 'Share These Results' link. Under 'Exact Match Found', the 'JOURNAL OF IMAGING' is listed with its publisher 'MDPI, MDPI AG, Grosspeteranlage 5, BASEL, Switzerland, CH-4052' and ISSN '2313-433X'. On the left, a 'Filters' sidebar shows 'Web of Science Coverage' and 'Core Collection' with checkboxes for SCIE, SSCI, AHCI, and ESCI, all of which are checked. A 'Find a Match' button is also visible in the sidebar. The bottom of the screen shows a Windows taskbar with various application icons and a system clock indicating 15:43 on 21-08-2025.

Web of Science

- **Effective Searching Techniques :**
 - Use quotes around phrases for precise search.
 - Add more keywords to refine results.
 - Utilize the "Refine Results" panel on the left for advanced filtering.
 - Explore options such as peer-reviewed journals, date, language, etc.
 - Broaden the search outcomes by taking into account alternative word forms, variations in word endings, and incorporating alternate search terms related to the research topic and connecting them with OR.
- **Example:** (for a structured approach) :
- **Research Topic:** Graph Theory and Network Analysis.
- **Search Criteria:** Title, Topic, Author.
- **Keywords:** Graph theory, network analysis, connectivity, algorithms.
- **Refine Results:** Select peer-reviewed journal articles, published within the last 3years.

Web of Science

- “Cited Reference Search” option enables a researcher to trace articles which have cited a formerly published paper.
- Using this element, it is possible to find how a familiar idea has been applied, improved, or extended subsequently.
- **Structured Search for Optimal Results**
- Structured searches enable narrowing and refining of results. Based on the re- searcher’s need the search result can be broadened or narrowed down using the built-in fields provided in the web of science website.
- This method is effective and ensures relevance and well-utilized time.
- When clicked on any of the search results, this website provides the title of the paper, authors, the type of journal, volume, issue number and year of publication, abstract, keywords, etc., so that the researcher has enough information to decide if it is worth while to acquire the fullversion of the paper.

Web of Science

- **“Example:**(for Structured Search):
- Objective: Recent developments in network connectivity algorithms.
- Structured Search: Use drop-down menus for precise selection (e.g.,Topic: Network Analysis, Sorting: Citations - Highest to Lowest).
- Outcome: Access a list of highly-cited articles on recent developments in network connectivity algorithms.

Google and Google Scholar

Google

- Google is a starting point for research.
- It is useful for finding freely available information from various sources such as reports from governments, organizations, companies, and so on, however there are limitations.

Limitations:

- It is a "Black box" of information, It searches everything on the Internet, with no quality control one does not know where results are coming from.
- There are limited search functionality and refinement options.

Google Scholar

- Focuses one's search to scholarly literature .However, there are limitations:
- Some of the results are not actually scholarly. An article may look scholarly at first glance, but is not a good source upon further inspection.

Google and Google Scholar

- Not comprehensive as some publishers do not make their content available to Google Scholar.
- There is limited search functionality and refinement options.
- **Search Operators in Google and Google Scholar**
- There are search operators in Google and Google Scholar that can be used to help narrow down the results. These help one find more relevant and useful sources of information. Here are some basic search operators:
- (i) OR- Broadens search by capturing synonyms or variant spellings of a concept.
Example: Synchronous OR asynchronous will find results that have either term present.
- (ii) Brackets/Parentheses () - Gathers OR'd synonyms together, while combining them with another concept.
Example: RAM (synchronous OR asynchronous).
- (iii) Quotation marks " "- Narrows the search by finding words together as a phrase instead of separately.
Example: RAM (synchronous OR asynchronous) "Texas Instruments".

Google and Google Scholar

- iv) Site - limits the search to results from a specific domain or website. This operator is helpful when searching specific websites such as the BC government.
- Example RAM(synchronous OR asynchronous) “Texas Instruments” site: <http://ieeexplore.ieee>.
- (v) File type- Limits the search to results with a specific file extension. One could look for pdf’s, PowerPoint presentations, Excel spreadsheets, and so on.
- Example: RAM (synchronous OR asynchronous) “Texas Instruments” site: <http://ieeexplore.ieee.org>, filetype: pdf.
- Search Tools and Other Options:
- The Search Tools button at the top of the Google results gives you a variety of other options, such as limiting the results by date.
- There are other operators and tools that one can use in Google and Google Scholar.
- Google is just one among many available search tools. Researchers are advised to consider other options beyond Google, for comprehensive research.

Effective Search : The Way Forward

- Scholarly and popular publications play distinct roles in sharing information in the field of engineering.
- A thorough understanding of their unique characteristics and knowing where to locate relevant information is essential for conducting research effectively.
- Scholarly Publications:
 - Scholarly publications are authored by researchers in a specific field of skill.
 - Such work cites all source contents used and is generally peer-reviewed for accuracy and validity before publication.
 - The audience for such works is fellow experts and students in the field. The content is typically more complex and advanced than those found in general magazines
 - Scholarly publications contribute to the academic knowledge base and are often a result of original research or in-depth analysis.
 - They adhere to a formal structure, including abstracts, methodologies, results, and conclusions.

Effective Search : The Way Forward

- Example:
- Title: "Advancements in Robotics: A Comprehensive Review"
- Authors: Dr. Engineer, Prof. Researcher
- Published in: Journal of Robotics
- Popular Publications :

Popular publications are informal and aim to reach a large number of readers, including experts and enthusiasts. They focus on news and trends, making them suitable for general reading.
- Popular publications serve as a bridge between experts and the general public, providing accessible information on current developments in the field. They may include interviews,
- case studies, and discussions that make the content more approachable

Effective Search : The Way Forward

- Example:
- . Title: "The Future of Sustainable Energy"
- . Author: Science Enthusiast
- . Published in: Engineering Today Magazine
- Search Strategies :
- Conducting a comprehensive search is essential for researchers. No single source provides all information needed, so various search tools should be utilized.
- Iterative Searching Process:
- Searching is an iterative process involving experimenting with keywords and operators.
- Evaluating results, and modifying searches. Once results are generated, carefully evaluate and assess them, utilizing filters to refine the relevant information.
- If the initial search doesn't yield the desired outcomes, be ready to modify your search terms based on initial results.
- Explore citations and references in relevant articles for further sources

Introduction to Technical Reading

- Technical reading is a vital skill for researchers, allowing them to remain well-informed and efficient in exploring research literature.
- Significance of Technical Reading in Research:
- For any active researcher, staying abreast(up-to-date) with the latest research findings in their field is now absolutely necessary.
- However, locating the right work to read can be challenging due to the vast and fragmented nature of the literature, with knowledge scattered across various sources.
- Knowing where to read is crucial, and relying on refereed journals and books from reputable publishers is preferred over easily accessible but potentially less reliable web articles.
- When reading an engineering research paper, the primary goal is to understand the technical contributions made by the authors.

Introduction to Technical Reading

- **Efficient Reading Strategies:**
- Given the abundance of journal articles, adopting a purposeful and efficient reading approach is crucial.
- Technical reading requires multiple readings, it is not the same as reading a newspaper.
- It often requires spending many hours on a single paper, needing multiple readings for a thorough understanding.
- Determine the worthiness of a paper through an initial Skimming to decide whether it is worth careful reading.
- **Time investment through Initial Skimming:**
- During this process, one typically glances through Title and Keywords, subheadings, introductory and concluding paragraphs, and any highlighted or emphasized text.
- The goal of initial skimming is to quickly assess the document's structure, main points, and relevance without delving into the details. It helps readers decide whether the paper merits a more in-depth and careful reading.

Introduction to Technical Reading

- **Effective Skimming Techniques:**
- When initiating the skimming process, begin with the title and keywords, as these elements are initial attention grabbers.
- One should then read the abstract to get an overview of the paper in minimum time
- If the abstract is of interest, one should skip most of the paper and go straight to the conclusions to find if the paper is relevant to the intended purpose.
- If found relevant, focus on reading the figures, tables, and captions, providing a broad understanding of the paper's content.
- If the paper has continued to be of interest, delve into the Introduction section to grasp the background information about the work and understand why the authors conducted that particular study and how it contributes to the field.
- The next sections to read are the Results and Discussion sections which is really the heart of the paper.
- One should really read further sections like the Experimental Setup/Modeling, etc., only if one is really interested and wishes to understand exactly what was done to better understand the meaning of the data and its interpretation

Conceptualising Research

- In formulating research objectives, the focus lies in its ability to centralize new knowledge, gaining acceptance and recognition within the research community.
- However, the journey begins with the conceptualization of the research itself.
- Besides being original and significant, a good research problem should also be solvable or achievable.
- This necessitates consideration of the methods and tools that can be used to obtain that new knowledge effectively.
- Significance, originality, and the theoretical framework, along with the methods and tools required to address a problem, usually come from the existing documented literature and knowledge in the field.
- For those engaged in Ph.D. level research or beyond, the task of conceptualizing the research becomes a personal responsibility.
- This step is particularly challenging as it demands an extensive knowledge of the literature in the field.

Critical and Creative Reading

- Reading a research paper is a critical process.
- The reader should not be under the assumption that reported results or arguments are correct.
- Rather, being suspicious and asking appropriate questions is in fact a good thing.
- **Asking the Right Questions:**
- The reader should ask the following questions during research paper analysis:
 - (i) Problem Solving: Have the authors attempted to solve the right problem?
 - (ii) Consideration of Alternatives: Are there simpler solutions that have not been considered?
 - (iii) Limitations: What are the limitations (both stated and ignored) of the solution?

Critical and Creative Reading

- **Asking the Right Questions:**
- (iv) Missing Links: Are there any missing links or gaps in the presented work?
- (v) Reasonable Assumptions: Are the assumptions made by the authors reasonable?
- (vi) Logical Flow: Is there a logical flow to the paper, or are there flaws in the reasoning?
- These questions need to be ascertained apart from assessing the relevance and importance of the work, through careful reading.

Critical and Creative Reading

- **Judgmental Approach:**
- Use of judgemental approach and boldness to make judgments is needed while reading.
- Flexibility to discard previous erroneous judgments is also critical for unbiased analysis.
- Correctness of Data: Ascertain whether the data presented in the paper is accurate and supports the argument.
- Gathering and Interpretation: Ascertain whether the data was gathered and interpreted in a correct manner.
- Alternative Datasets: Decipher whether an alternative dataset would have been more compelling

Taking Notes While Reading

- Effective reading is essential for good writing in research. The process of taking notes during and after reading serves as a bridge between reading and writing.
- **Importance of Note-Taking:**
- **Building Knowledge:** Notes aid in building on acquired knowledge.
- **Memory Aid:** There is a well-known saying that the faintest writing is better than the best memory, and this holds true for researchers who need to read and build on acquired knowledge.
- **Long-Term Benefits :** Efforts in note-taking become particularly significant when revisiting and rereading the material after a long time.
- **Note-Taking methods:**
- **Marginal Notes:** Many researchers take notes on the margins of their copies of papers or even digitally(annotate) on an article.
- **Contents to Highlight:** In each research paper a lot of things like Definitions, explanations, key concepts, questions, and criticisms will be highlighted.

Concluding a Thorough Reading

- **summary:** On completing a thorough reading, a good technical reading should end with a summary of the paper in a few sentences describing the paper's contributions
- **Elucidating Technical Merit:** Elucidate on the importance of understanding the technical merit by comparing the paper with existing works in the same area.
- **Evaluation of Innovation:** Assess whether the paper introduces new ideas or implements existing ones in innovative ways.
- **Novel Framework:** Evaluate if the paper introduces new ideas, or implements existing ideas through experiments or in new applications in new ways, or consolidates different ideas under a novel framework.
- **Determining Contribution through Comparative Analysis :** The true contribution of a paper is better understood by considering other papers in the same area. Research students should thoroughly analyze how the paper stands out when compared to existing works in the field.

Attributions and Citations

- Academic writing, by definition, must follow certain rules and conventions.
- Among the most important of these are the rules and conventions about citing, referencing, attributing, and acknowledging the works of others. That means giving proper credit wherever due.
- **Citing:** Citing is the practice of quoting from, referring to other authors' works and ideas in the text of our work in such a way that the context is clear to the reader.
- **Referencing:** Referencing is the listing of the full publication details of a published work that is cited so as to give background information to the readers.
- **Attribution:** Properly crediting the contributors for their contributions
- **Acknowledgment:** Acknowledgment in research publications indicates contributions to scientific work.

Attributions and Citations

- Acknowledgment, attributions, and citations differ in the manner of their application.
- Acknowledgment is arguably more personal, singular, and simply an expression of appreciations and contribution.
- **Citations: Functions and Attributes**
- **Role of Citations (References):**
- Citations credit others for their work, providing readers the means to trace the source publication if necessary.
- In any new document, any borrowed portion of someone else's work or ideas, be it in papers, patents, or presentations, must be clearly cited to avoid plagiarism.
- This rule applies to all written sources, encompassing texts, images, sounds, etc. and failure to do may be considered plagiarism.

Attributions and Citations

- Depending on the exact type of material, the researcher may need to give due credit to the creator of the original source.
- The growth of knowledge in any field of study, especially in technological fields, is primarily incremental and a researcher invariably and naturally builds upon prior information.
- It is true that research needs to leverage the prior art around research interest to make further development, at the same time it is important to ensure that credit for that existing knowledge is suitably acknowledged.
- When a bibliography of previously published patents or papers is placed in the new works of a researcher, a connection is established between the new and previous work.
- As per relevance to context, the researcher provides due credit using a citation.

Attributions and Citations

- Citations help the readers to verify the quality and importance of the new work and justification of the findings.
- It is a way to tell readers that certain material in the researcher's present work has come from another source and as an ethical responsibility, appropriate credit has been given to the original author or writer.
- Materials that can be cited include journal papers, conference proceeding, books, theses, newspaper articles, websites, or other online resources and personal communication.
- Preferably, citations should be given at the end of a sentence or the end of a paragraph as can be seen even in this paragraph.
- Citation must contain enough details so that readers can easily find the referenced material

Attributions and Citations

- **A researcher needs to cite each source twice:**
- In-text citation, in the text of the article exactly where the source is quoted or paraphrased.
- A second time in the references, typically at the end of the chapter or a book or at the end of a research article.
- Most citation styles have the same or similar elements, but differ on the order of elements and layout.
- Unless otherwise specifically required by a particular journal or a book, one may choose any style of one's choice if one is consistent.
- The citation elements differ and so what is to be recorded can differ from one source to another.
- It is also important to mention the date the source was published and sometimes also the date it was accessed by the researcher if it is related to web content.

Attributions and Citations

- The citation elements differ and so what is to be recorded can differ from one source to another.
- It is also important to mention the date the source was published and sometimes also the date it was accessed by the researcher if it is related to web content.
- LaTeX, a document preparation system often used by engineering researchers to automatically format documents that comply with standard formatting needs, is very effective to track and update citations.
- LaTeX has a steep learning curve and will be repeatedly used in this book to address different issues pertaining to technical writing which is intimately linked with research for engineers.

Attributions and Citations

- There are three main functions of citation:
- Verification function: Authors have a scope for finding intentional or unintentional distortion of research or misleading statements. Citation offers the readers a chance to ascertain if the original source is justified or not, and if that assertion is properly described in the present work
- Acknowledgment function: Researchers primarily receive credit for their work through citations. Citations play crucial role in promotion of individual researchers and their continued employment. Many reputed organizations and institutes provide research funding based on the reputations of the researchers. Citations help all researchers to enhance their reputation and provide detailed background of the research work.

Attributions and Citations

- Documentation function: Citations are also used to document scientific concepts and historical progress of any technology over the years. Citations are the currency that authors would wish to accumulate and the technical community gives them credit for these contributions.
- When other authors make citations, they honour those who initiated the ideas.
- Authors demonstrate their comprehension skills by identifying, estimating, and incorporating other's research work and then create and express their own ideas precisely while acknowledging ownership of ideas through citation.

Attributions and Citations

- There are certain cases when references do not fulfill the actual goal of citations and acknowledgments, and thus do not benefit the reader
- 1. Spurious citations:
- In certain cases, when citation is not required or an appropriate one is not found, if the author nevertheless goes ahead with including one anyways, it would be considered as a spurious citation.
- These sorts of citations do not add any value to the reader in terms of properly understanding the paper. Such actions result in loss of time of the reader or reviewer in looking for the cited paper that is otherwise not relevant.
- Just as due credit should be given to a paper through citation, inappropriate credit must be avoided so that the credibility of a research work or of the journal or conference proceedings where that paper is published is not lost through this sort of carelessness.

Attributions and Citations

- **2. Biased citations:** When authors cite the work of their friends or colleagues despite there being no significant connection between the two works, or when they do not cite work of genuine significance because they do not wish to give credit in the form of citation to certain individuals, then such actions can be classified as biased citations.
- **Self-citations:** There is nothing wrong in citing one's prior work if the citation is relevant.
- Self-citation of prior papers is natural because the latest paper is often a part of a larger research project which is ongoing.
- Sometimes, it is also advantageous for the reader because citations of all the related works of the same author are given in one paper and this may reduce the effort of the reader in trying to find the full versions of those papers.

Attributions and Citations

- Coercive citations: Despite their shortcomings, impact factors remain a primary method of quantifying research.
- One side effect is the creation of an incentive for editors to engage in coercion, pressuring authors to add citations to the editor's journal.
- Even if not explicitly stated, the implied message is that the author could either add citations or risk rejection.
- Such coercive demands consequently diminish the reputation of the journal.

Attributions and Citations

- However, it is helpful and ethical only if all the papers are relevant to the present work.
- However, there can also be negative impact on the journal as well as individual researchers due to inappropriate and irrelevant self-citations.
- Self-citations in such cases may be either spurious or biased or even both.
- Editors of journals who ignore such types of citations and allow by negligence or otherwise, to be included in published materials end up directly or indirectly altering the impact factor of those publications.
- An Impact Factor (IF) is a quantitative measure of a journal's influence, reflecting how often its articles are cited by other publications over a specific period, typically two years. Calculated annually by Clarivate.

Attributions and Citations

Journal Impact Factor Calculation

$$\begin{array}{l} \text{2017} \\ \text{Journal} \\ \text{Impact} \\ \text{Factor} \end{array} = \frac{214}{96} = 2.229$$

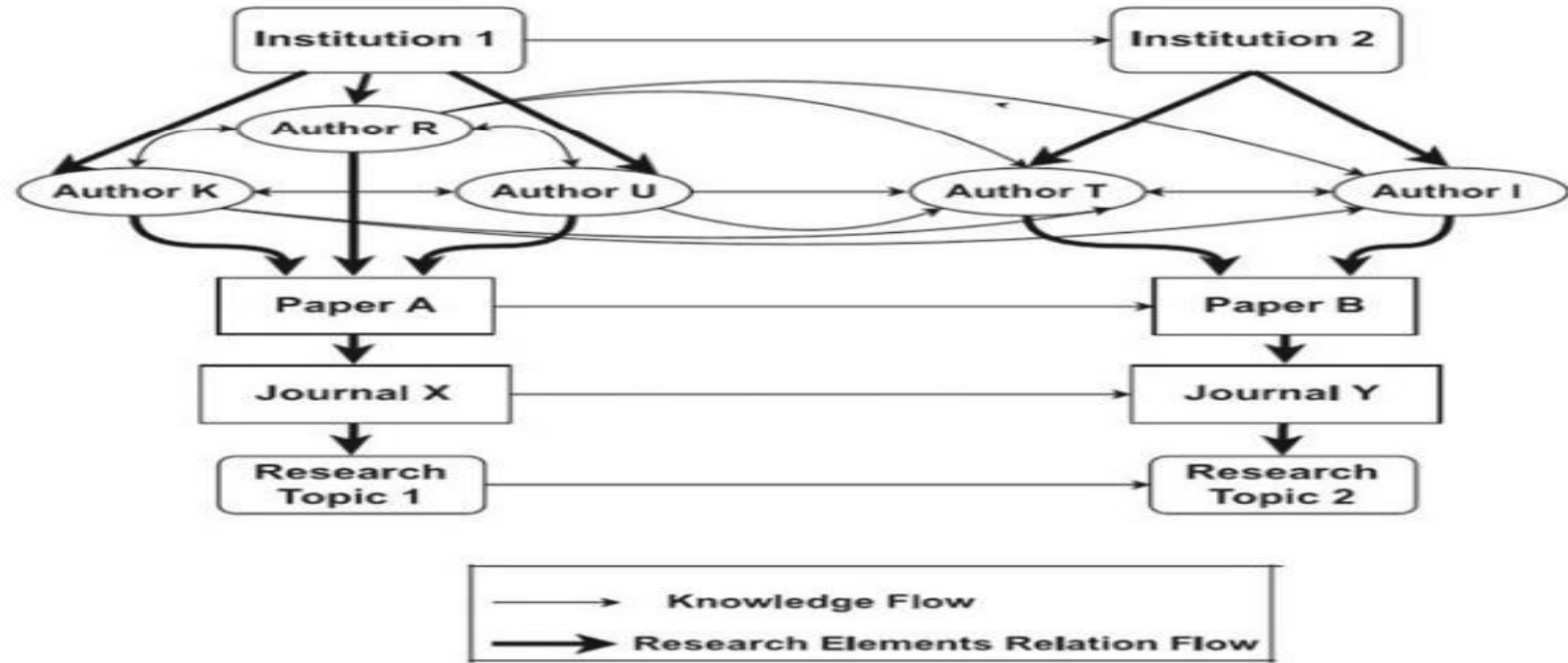
How is Journal Impact Factor Calculated?

$$\text{JIF} = \frac{\begin{array}{c} \text{Citations in 2017 to} \\ \text{items published in} \\ \text{2015 (109) + 2016} \\ \text{(105)} \end{array}}{\begin{array}{c} \text{Number of citable} \\ \text{items in 2015 (46) +} \\ \text{2016 (50)} \end{array}} = \frac{214}{96}$$

Knowledge Flow through Citation

- Knowledge flows through verbal communications, books, documents, video, audio, and images, which plays a powerful role in research community in promoting the formulation of new knowledge.
- In engineering research, knowledge flow is primarily in the form of books, thesis, articles, patents, and reports. Citing a source is important for transmission of knowledge from previous work to an innovation.
- Production of knowledge can be related to the citation network. Knowledge flow happens between co-authors during research collaboration, among other researchers through their paper citation network, and between institutions, departments, research fields or topics, and elements of research.

Knowledge Flow through Citation



Illustrative Relationship:

The above figure shows the relationship between citations, knowledge flow, and elements such as researchers, papers, journal publications or conferences, and institutions.

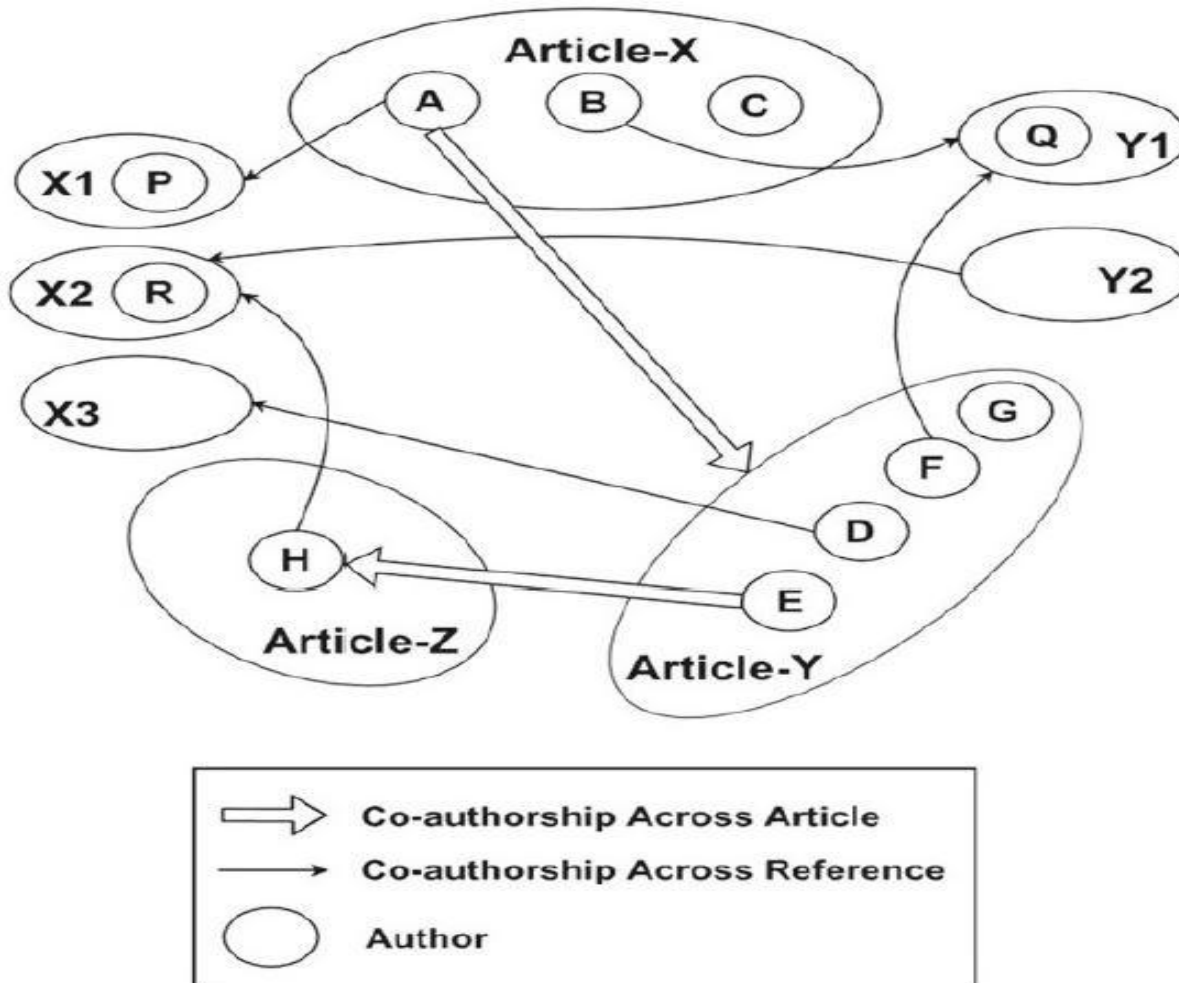
If paper A is cited by paper B, then knowledge flows through citation networks across institutions.

Knowledge Flow through Citation

Interdisciplinary Nature of Research :

- The complex interdisciplinary nature of research encourages scholars to cooperate with each other to grab more advantages through collaboration, thereby improving quality of the research.
- Soorya Moorthy, examined the citation impact of the South African publications among different collaboration types, discipline and sectors, and observed that co-authored publications had more citations than single author paper
- There was a positive co-relation between number of authors and the number of citations.
- Following figure shows a relationship between co-authorship and different types of citations

Knowledge Flow through Citation



- Three articles (X, Y, and Z) and five references (X1, X2, X3, Y1, and Y2) of article X and Y, respectively, are considered.

- A, B, and C are authors of article X, and D, E, F, G, and also A are authors of article Y. Article Z has two authors H and E.

- References

X1, X2, X3, Y1, and Y2 have authors (A, P), (H, R), (D), (Q, B, F), and (R), respectively.

Knowledge Flow through Citation

- Based on co-authorship citation network,
 - references X1 and Y1 are considered self-citation,
 - reference X3 is a level-1 co-author citation because author of article Y is direct collaborator of author A
 - reference X2 is a level-1 co-author network because author A is collaborator of E who collaborated with H.
- We conclude that papers which frequently cite collaborators will also often cite collaborators of collaborators. Collaborations certainly impact citation counts.

Citing Datasets

- Engineering research relies heavily on data. Data citations credit dataset creators, ensuring legal attribution and aiding in future retrieval.

Ownership of Data :

- Ascertaining the ownership of data can be a complicated issue especially with large datasets, and issues of funding can also make it a difficult matter.
- A researcher should obtain necessary permission for using data from a particular source.

Citation Information:

- Citations related to datasets should include enough information so that a reader could find the same dataset again in the future, even if the link provided no longer works.
- It is proper to include a mixture of general and specific information to enable a reader to be certain that the search result is the same dataset that was sought.

Citing Datasets

Examples:

1. Historical Data, Sotavento (Wind Farm), Corunna, Spain (July 2016): [Accessed: 4 Oct, 2016] Retrieved from <http://www.sotaventogalicia.com/en/real-time-data/historical>
2. Deb, D (2016). [Personnel survey]. Unpublished raw data.

Styles for Citations :

Citation styles differ primarily in the order, and syntax of information about references, depending on difference in priorities attributed to concision, readability, dates, authors, and publications.

Some of the most common styles for citation (as well as other aspects of technical writing) used by engineers are as follows:

Citing Datasets

ASCE Style(American Society of Civil Engineers):

- **(a) Reference list:**
- This part is to be placed in the bibliography or references at the end of the article or report. A template with example for the same is given below:
- **Template for books:**
- Author Surname, Author Initial. (Year). Title: Publisher, City, Pages Used.
- Example: Wearstler, K., and Bogart, J. (2004). Modern glamour. Regan Books, NY.
- **Template for Websites :**
- Author Credentials and Company Name (Year). 'Title'. [Website URL (Accessed: Date)].
- Example: Blade cleaning services (2015): [Website URL (Accessed: Oct. 29, 2016)].

Citing Datasets

- **Template for Journal Publications :**
- Author Surname, Author Initial. (Year). 'Title'. Publication Title, Volume (issue), Pages.
- Example: Johnston, L. (2014). "Inconvenient Truth Expanded The Climate Change Dialogue". 1–160.
- **IEEE Style (Institute of Electrical and Electronics Engineers):**
- IEEE style is standard for all IEEE journals and magazines, and is frequently used for papers and articles in the fields of electrical engineering and computer science.
- The IEEE style requires endnotes and that references be cited numerically in the text.
- Those submitting to an IEEE publication should see guidelines for the specific journal or magazine and may also refer to the complete IEEE editorial style manual. Some examples of IEEE styles of citations for different types of sources are enumerated below:
- **Chapter in an edited book example:**
- [1] A. Rezi and M. Allam, "Techniques in array processing by means of transformations," in Control and Dynamic Systems, Vol. 69, multidimensional Systems, C. T. Leondes, Ed. San Diego: Academic Press, 1995, pp. 133-180.

Impact of Title and Keywords on Citations

- The citation rate of a research paper is influenced by various factors, including journal significance, publication types, research area, and the importance of the published work.
- Additionally, factors such as title length, type, and selected keywords impact citation count
- **Significance of Research Paper Titles:**
- The title is the most important attribute of any research paper.
- It is the main indication of the research area or subject and is used by researcher as a source of information during literature survey.
- Title plays important role in marketing and makes research papers traceable.
- A good title is informative, effectively represents the paper to readers, and gains their attention.
- While some titles may be informative, they might not engage readers, and some titles might be attractive but not informative or related to the readers' research area.

Impact of Title and Keywords on Citations

- The download count and citation of a research paper can be influenced by its title.
- There are three different aspects which provide a particular behavior to the title:
 - (i) types of the title,
 - (ii) length of the title, and
 - (iii) presence of specific markers.
- The title is a crucial attribute of any research paper, influencing its traceability and attention from readers.

Studies on Title Characteristics :

- Research by Stremersch et al. analyzed papers published between 1990 and 2002, in the area of research and studied relationship between title characteristics and citation, which concluded that title length positively affects the number of citations.

Impact of Title and Keywords on Citations

Studies on Title Characteristics :

- In another study, Sagi and Yechiam, found that highly amusing titles have fewer citations and pleasant titles have no significant relation to citation counts.
- Another study by Jacques and Sebire, observed a strong association between title length and citation rates, with highly cited articles having more than twice as many words in the title compared to lower cited papers
- Jamali and Nikzad analyzed several open access papers and found that articles with question-type titles are downloaded more but cited less compared to the descriptive or declarative titles.
- Analysis by Habibzadeh and Yadollahie also indicated a strong association between longer titles and higher citation rates. Longer titles mainly include the study methodology and/or results in more detail, and so attracts more attention and citations.

Impact of Title and Keywords on Citations

- **General Observations :**
- In general, titles containing a question mark, colon, or reference to a specific geographical region tend to be associated with lower citation rates.
- Titles that describe results typically receive more citations than those describing methods.
- Additionally, review articles and original articles usually receive more citations than short communication articles.
- Including at least two keywords in the title can increase the chance of finding and reading the article as well as get more citations.

Impact of Title and Keywords on Citations

Significance of Keywords in Research Articles :

- Keywords represent essential information as well as main content of the article, which are relevant to the area of research.
- Search engines, journal, digital libraries, and indexing services use keywords for categorization of the research topic and to direct the work to the relevant audience.
- Keywords are important to ensure that readers are aware about research articles and their content.
- If maximum number of allowable keywords are used, then the chance of the article being found increases and so does the probability of citation count of the article.
- Usage of new keywords should be minimal as such keywords may not be well known to the research community and so may lead to low visibility of the article.